

MOHAMMED SAHAL SHAIKH

Graduate Mechanical Engineer

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South African Permanent Resident | Code 10 driver's licence | Available immediately

PROFESSIONAL SUMMARY

Mechanical Engineering (Honours) graduate from Curtin University, Australia (SAQA-evaluated, NQF level 8), with a 72.89% weighted average and a 78% final year. Honours thesis cross-validated a porous-media model by image analysis, ANSYS Fluent CFD and physical testing, with CFD agreeing to within 0.17%. Designed, built and instrumented a solar-panel test rig whose findings improved panel efficiency by 10.6%; completed GCAA-approved aircraft maintenance training; co-authored peer-reviewed research in additive manufacturing (Springer Nature). Founded four university student chapters and led competitions spanning four countries and 12 universities. Seeking a graduate mechanical, design, simulation, energy or project engineering role in Gauteng.

Languages: English, Afrikaans and Urdu (fluent); Turkish (conversational).

EDUCATION

BEng (Honours), Mechanical Engineering | Curtin University (Australia), Dubai campus 2022 – 2026

- Weighted average **72.89%**; final-year average **78%**; FEA and CFD units both above **80%**; Upper Second Class Honours.
- Washington Accord-recognised degree; **SAQA-evaluated at NQF level 8**.
- Honours thesis: cross-validated a porous-media model of eggplant tissue by image analysis (**4,498 pores**), ANSYS Fluent CFD (**within 0.17%**) and gravimetric testing (**96.6%** of theoretical uptake).

National Senior Certificate (Matric), Bachelor's pass | Ahmed Timol Secondary School, Azaadville 2019

- Average **77%** with distinctions in Physical Sciences (**80%**), Accounting (**80%**) and Life Orientation (88%); Afrikaans FAL 76%, English HL 75%, Life Sciences 74%, Mathematics 67%.

TECHNICAL SKILLS

CAD & Design: SolidWorks (incl. Simulation), Autodesk Inventor, AutoCAD, Onshape, FreeCAD; engineering drawings, GD&T, tolerance analysis, BOMs, DFM/DFA

Simulation & Analysis: ANSYS Fluent, ANSYS Mechanical, MATLAB, Simulink, SimScale; CFD, FEA, heat transfer, fatigue and Weibull reliability, FMEA, root cause analysis

Programming & Data: Python (OpenCV, pandas, scikit-learn), C++ (basic), Arduino (C/C++), Git, Advanced Excel, LaTeX, LTspice

Energy & Sustainability: Solar PV testing, lifecycle assessment (openLCA), process modelling (DWSIM), techno-economic analysis, Ashby material selection

Test & Manufacturing: Test-rig design and build, instrumentation and data acquisition, sheet metal, 3D printing, laser cutting, soldering

Project & Professional: MS Project, MS Office, project scheduling, risk registers, cost estimation, stakeholder management, technical report writing

PROFESSIONAL EXPERIENCE

Transnational Academic Group (operator of Curtin University Dubai, UAE): rehired for three roles, 2023 – 2026; offered a permanent position on completion (declined).

Solar Energy Research Intern | Transnational Academic Group, Dubai Feb 2026 – Jun 2026

- Designed in SolidWorks, fabricated and instrumented a solar-panel soiling test rig, and developed the method used to quantify dust-driven losses in PV efficiency and service life.
- Ran controlled soiling tests with supporting CFD analysis; the resulting cleaning and maintenance schedule **improved panel efficiency by 10.6%**.
- Synthesised 20+ peer-reviewed studies of degradation mechanisms into technical reports.

Project Coordinator, ICISAS 2025 Conference | Transnational Academic Group, Dubai Dec 2024 – Mar 2025

- Led the peer-review operation: screened and triaged **450+ submissions** for scope and quality, then assigned each to reviewers from a **1,000+ reviewer** pool by technical expertise.
- Managed the coordination team and the end-to-end submission system to deliver reviews to the conference timeline.

Materials & Solar Research Intern | Transnational Academic Group, Dubai Aug 2023 – Mar 2024

- Compared **5 solar-panel technologies** on performance, degradation and end-of-life behaviour.
- Evaluated 2 aluminium-alloy combinations, analysing how selective laser melting affects microstructure and mechanical properties; co-authored the resulting **Springer Nature** publication.

- Taught CAD and robotics to **~80 students across 6 schools** and coached a FIRST LEGO League team to the **national finals**.

Aviation Maintenance Training Placement | Sarsan Aviation Academy (GCAA CAR-147), Dubai

Aug 2023

- Completed supervised practical training on a Piper PA-28 across **26 aircraft components**: sheet-metal repair, safety wiring, wing attachment and flight-control rigging, plus gas turbines, avionics and aviation regulations.

PROJECTS

Industrial gearbox design (1.6 kW, 8-year life, <1% failure probability, full SolidWorks drawing package), residential renewable microgrid, sustainable footwear manufacturing process, search-and-rescue robot and more. Full write-ups, drawings and the Honours thesis at msahal31100-dot.github.io/portfolio.

LEADERSHIP

- **Founder** of four student chapters at Curtin University Dubai: IEEE, ASHRAE, Robotics Club and the Federation of Economic Development Association (FEDA).
- **President, IEEE Student Branch** (2025 – 2026): led the inaugural IEEE Curtin ASPIRE SDG Innovation Challenge across **4 countries** (UAE, Sri Lanka, Malaysia, Australia); ran IEEE Tech Fest and co-led IEEE Convergence with Curtin Colombo, including its first Simulation Engineering event.
- **President, ASHRAE Student Branch** (2025 – 2026): led the ASHRAE Poster Competition for three years; the 2025 edition drew **40+ entries from 12 UAE universities**.
- **Lead, PMI UAE Student Club** (2025 – 2026): organised a career-readiness event for **50–60 participants**, owning budget, speakers and logistics; partnered with clubs in Jordan, Lebanon and Saudi Arabia.
- **Treasurer, Engineers Australia Student Chapter** (2023 – 2026): managed event budgets and finances across a three-year term.
- **Also**: Events Officer, ASME Student Chapter; Engineering School Officer, Student Council; three years mentoring school STEM challenge teams.

PUBLICATIONS

- Gadher, S., **Shaikh, M.S.M.S.**, Selvan, C.P. (2025). Effects on the Microstructure and Mechanical Properties of the AlSi14 Alloy Produced by the Selective Laser Melting (SLM) Technique. ICISAS 2023, Springer Nature. doi:10.1007/978-3-031-68952-9_15
- Co-author of four further peer-reviewed papers: MARLOG 15 (2026) on zero-emission logistics corridors, and three IEEE ASET 2025 papers in signal processing and data systems. Full list on LinkedIn.

AWARDS

- **1st Place**, Mubarmij50 Robotics Competition (2024), leading the Curtin Robotics Club team.
- **1st Place**, Manipal University Dubai Hydrojet Competition (2022).
- **1st Runner-Up**, Western Australia Government Climate Action Challenge (2024), with an official WA Government recognition letter.
- **National Finalist**, James Dyson Award (2023): led design of a solar cooking backpack that cooked a meal in ~2 hours, 4 hours faster than traditional solar cookers.
- **Winner**, ASHRAE Falcon Debate, Occupant Wellbeing team.

CERTIFICATIONS & TRAINING

- Ansys (2026): Supersonic, Subsonic and Two-Dimensional Flows; Shock-Expansion Theory; Flow over an Airfoil.
- ASME: GD&T Drawing Interpretation; Design for Sustainability; Industry 4.0 (two modules); Digital Literacy and Data Skills.
- Royal Aeronautical Society & Vernewell Academy: 5-week aerospace mentorship (space law, contracts, commercial space operations).

REFERENCE

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